

Exercise 6: Library Management System

1. Understand Search Algorithms

Linear Search : Checks each book one by one until the target is found.

Binary Search : Repeatedly divides the sorted list into halves to locate the target.

2. Setup

```
class Library {  
    int bookId;  
    String title;  
    String author;  
  
    Library(int bookId, String title, String author) {  
        this.bookId = bookId;  
        this.title = title;  
        this.author = author;  
    }  
}
```

3. Implementation

```
class Search {  
    public void linearSearch(Library[] books, String title) {  
        for(int i = 0; i < books.length; i++) {  
            if (books[i].title.equals(title)) {  
                System.out.println(title + " Book Found");  
                return;  
            }  
        }  
        System.out.println("Book Not Found");  
    }  
  
    public void binarySearch(Library[] books, String title) {  
        int left = 0, right = books.length - 1;  
        while(left <= right) {  
            int mid = left + (right - left) / 2;  
            int res = books[mid].title.compareTo(title);  
            if (res == 0) {  
                System.out.println(title + " Book found");  
                return;  
            } else if (res < 0) left = mid + 1;  
            else right = mid - 1;  
        }  
        System.out.println("Book Not Found");  
    }  
}
```

```
}
```

```
public class LibraryManagementSystem {  
    public static void main(String[] args) {  
        Library[] books = {  
            new Library(101, "Algorithms", "CLRS"),  
            new Library(102, "Database Systems", "Korth"),  
            new Library(103, "Java Programming", "Herbert Schildt"),  
            new Library(104, "Operating Systems", "Galvin"),  
            new Library(105, "Python Basics", "Guido van Rossum")  
        };  
    }  
}
```

```
Search search = new Search();
```

```
System.out.println("Linear Search:");  
search.linearSearch(books, "Java Programming");
```

```
System.out.println("\nSearching for a non-existing book:");  
search.binarySearch(books, "Computer Networks");  
}  
}
```

4. Analysis

<i>Operation</i>	<i>Best Case</i>	<i>Worst Case</i>	<i>Average Case</i>
Linear Search	$O(1)$	$O(n)$	$O(n)$
Binary Search	$O(1)$	$O(\log n)$	$O(\log n)$

Linear Search

- Small datasets.
- Unsorted data.
- Simpler implementation.

Binary Search

- Large datasets.
- Sorted data.
- Faster search performance.